

Meeting Summaries of Sep 11

Date: September 11

Time: 1:00–1:25 PM(PDT)

Attendees: Prof. Jan, Dr. Yue and Feng

Topic: Feng reporting his work on community species composition and species assemblage analysis.

Summary of Discussion

- Feng presented his imputation work on water velocity data, using the 18 DR sites to impute the unknown Lake St. Clair and St. Clair River water velocity data. Some background info was used to guide the imputation.
- Prof. Jan suggested Feng to use the complete velocity data of Detroit River when it comes to velocity imputation. There are 213 DR sites with water velocity values in environmental data. And be careful in figuring out whether it is imputating near-bottom or near-surface velocity, since the two types of velocity can be quite different to sediment species.
- Feng reported the reproduction of using *Ward's* method with euclidean distance to cluster the sites, based on the *octave* transformed species abundance. This is a *Q-Mode* that analyzes the patterns among the sites (rows).
- Feng also presented the *R-Mode* analysis that clusters the species (columns) based on their correlation in variation across the sites regardless of other habitat features. *K-means* and *Kendall's W* coefficient were used to do this species assemblage analysis, following the steps on page 125-128 of *Numerical Ecology with R*.

Action Items

- Feng needs to check and correct the imputed water velocity data if it goes into next analysis. That could be :
 - Use the total 213 DR sites with velocity data ¹ to impute the missing LSC and SCR velocity.
 - Check the relevant articles about the near-bottom velocity comparisons among LSC, SCR and DR, to make sure the imputation is reasonable to some extent.
- Feng to proceed with the Habitat-Community Discriminant Analysis, which requires LDA to find best linear combinations of habitat features that separate the community types identified from *Q-Mode* clustering analysis.

Follow-Up

Temporary meeting time: Dec 18, 1:00 - 1:25 PM (PDT)

¹From Zhang's thesis, the DR velocity data was estimated for near-bottom, so the imputation should also be for near-bottom for the other areas.

Supportive materials

- A meeting recording has been stored at Feng's laptop.